Shun Akiyama

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Education

University of British Columbia — **Expected Graduation: May 2029**
Bachelor of Applied Science — *Cumulative GPA: 93.6/100*

- Relevant Coursework: Software Construction, Computing Systems I & II, Algorithms & Data Structures, Circuit Analysis, Linear Algebra, Multivariable Calculus, Differential Equations, Probability

Experience

UBC BizTech — **Apr 2026 – Present**
Software Engineer — *Vancouver, BC*

- Building a full-stack hackathon and conference platform for UBC's largest technology club, serving 1,900+ users and streamlining event registration, membership, and payment workflows.
- Integrated AWS API Gateway, Lambda, Cognito, and DynamoDB across a 20-microservice architecture to enable secure authentication, scalable backend APIs, and event registration logic.
- Implemented Stripe webhook automation for $20,000+ in membership payments, reducing manual payment verification and triggering event confirmation workflows.
- Partnered with product, design, and engineering teams through GitHub-based development, pull requests, code reviews, and sprint planning to ship reliable platform features.

UBC Formula Electric — **Sep 2025 – Present**
Software / Firmware Engineer — *Vancouver, BC*

- Developing embedded firmware in C/C++ for STM32 microcontrollers in a Formula SAE electric vehicle.
- Configuring STM32 peripherals, including ADC, TIM, I2C, SPI, and GPIO, using CubeMX and initializing them in the hardware layer.
- Integrating brake pressure sensors, suspension systems, tire temperature sensors, IMUs, and coolant flowmeters into the rear sensor module by developing functions to retrieve data from the hardware, calculate quantities using datasheets, and communicate over the CAN bus.
- Validating firmware using Segger J-Link to flash PCBs and conducting Software-In-Loop testing for CAN message transmission
- Constructed Chimera, a Python abstraction script for the electrical team to test their hardware and object-oriented drivers for potentiometer and software watchdog implementation
- Collaborated effectively with engineers across various disciplines, practicing communication and leadership skills to support vehicle design, testing, and competition readiness.

Thompson Community Centre — **Sep 2024 – Nov 2025**
Badminton Instructor — *Richmond, BC*

- Instructed over 100 beginner students ages 6 - 13 years old, emphasizing proper technique, footwork, and sportsmanship.
- Managed weekly classes of up to 13 students and developed engaging lesson plans to support individual and team growth.
- Collaborated with staff to host mini in-house tournaments to engage students in a competitive environment.

Projects

Stock Analyzer | *Python, NumPy, pandas, Matplotlib* — **February 2026**

- Built a Python-based financial data pipeline using NumPy and pandas to process historical stock data and compute time-series features
- Implemented core financial metrics including simple/log returns and rolling volatility with correct time alignment and NaN handling
- Visualized price, return, and volatility dynamics using Matplotlib, identifying key behaviors such as mean-reverting returns and volatility clustering

Autonomous Pick-Up Claw | *Arduino, C++* — **January 2026**

- Developed an autonomous claw using an Arduino kit, servo motors, sonar sensor, and aluminum sheet metal.
- Implemented an Exponential Moving Average Filter and a Median Filter to reduce noise in data points for sonar sensing distance

Technical Skills

Languages: C/C++, Python, MATLAB
Developer Tools: GitHub, VS Code, CubeMX, Segger J-Link
Libraries: NumPy, pandas, Matplotlib, STM32 HAL